

Optical Fibre Access Technologies

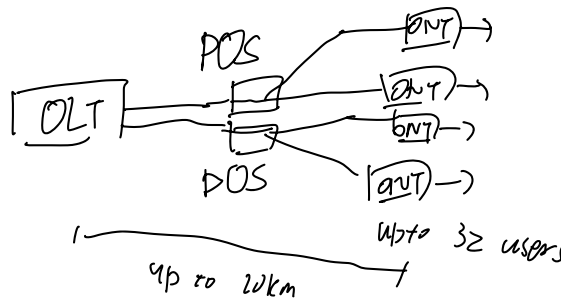
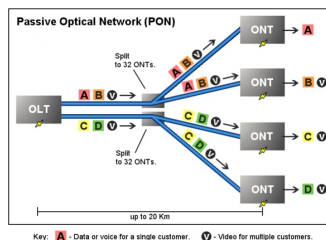
Approaches to deployment (manages the optical fibers to the home)

- **Full Fibre** - where a fibre is directly connected to home.
- **Fibre to the Cabinet** - fibre is connected to the grey box on the street and coaxial electrical cable to the home.

Passive optical Network

PON is a fiber-optic technology used for delivering high-speed internet, voice, and video services to residential and business properties without the use of switches or amplifiers. It uses only unpowered devices to carry signals.

高速, 互联网等, 到用户端 (居民楼, 办公室), 无源。



Optical line terminal

An optical line termination (OLT) is a device which serves as the service provider endpoint of a PON. 光线路终端(OLT), 服务提供商 (ISP) 端点的设备。

Passive optical splitter

The splitter divides the optical signal into multiple outputs without requiring any external power. It allows a single fiber from the OLT to serve multiple endpoints (up to 32 clients). 分岔信号, 无源, 从OLT到ONT, 多达32位用户。

Optical network terminal

An ONT (Optical Network Terminal) is the device that serves as the customers endpoint of a PON. ONT光网络终端 (光猫)。PON 用户端设备。

The ONT/ONU is the device that exists at user's home or office. ONT/ONU 是用户家庭或办公室中的设备。

Types of PON

GPON: maximum user capacity is 1Gbps.
XGPON: maximum user capacity is 10Gbps.

Best Networking protocol for PON

Ethernet.

The Ethernet protocol works by dividing up a stream of data into shorter pieces called **frames**. Each frame contains information about the source and destination addresses (**MAC address**) and **IP** addresses. They can be read and used to subsequently **direct** signals at switches to control the data's movement through the network.

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